

Martin Mirchev

PhD Student, National University of Singapore

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PhD student at the National University of Singapore, working in the Programming Languages and Software Engineering (PLSE) Lab within the Trustworthy Software Systems group, advised by Professor Abhik Roychoudhury. Research spans automated program repair, automated verified evolution of software systems, and AI-assisted software security.

Education

PhD in Computer Science

2022 – 2028 (*expected*)

National University of Singapore (*qualifying examination passed*)

B.Sc. in Computer Science and Engineering

2019 – 2022

Eindhoven University of Technology (*GPA 8.1/10, cum laude*)

Work Experience

Research Consultant

2026 – present

Sonar

- Continuing research collaboration on automated program repair and AI-assisted software engineering

Teaching Assistant

Jan–Apr 2026, Sem. 2 AY2025/2026

National University of Singapore, School of Computing

- CS3234 Logic for Proofs and Programs

Research Engineer

Feb 2025 – 2025

Sonar

- Worked on agentic program repair and AI-assisted issue resolution, including SonarQube-integrated tooling, following Sonar's acquisition of AutoCodeRover

Co-Founder & CTO

2024 – Feb 2025

AutoCodeRover

- NUS spin-off building an autonomous AI agent platform for automated software development, combining LLMs with code search and reasoning to automate parts of the software development lifecycle
- Acquired by [Sonar](#) in February 2025

Teaching Assistant

Jan–Apr 2023, Sem. 2 AY2022/2023

National University of Singapore, School of Computing

- CS3213 Foundations of Software Engineering

Teaching Assistant

Nov 2020 – May 2022

Eindhoven University of Technology

- Courses: 2ITB0 Provable Programming, 2IT80 Discrete Structures, 2IT60S Logic and Set Theory, 2IT90 Automata, Language Theory and Complexity

ANTLR Software Consultant

Jul 2020 – Aug 2022

Kruggle

- Migrated existing grammars from ANTLRv3 to ANTLRv4; investigated performance and memory issues
- Added support for new grammars: Scala, Swift, Kotlin, Objective-C, Go, COBOL, ABAP

- Added internationalization capabilities and helped maintain the MyPavel project
- Developed software for BeagleBone; extended the Prototizer tool to generate Node.js/Ionic projects with PostgreSQL schemas
- Worked on multiple Zend/PHP backends

Publications

- M. Mirchev, R. Shariffdeen, H. Ruan, Y. Zhang, A. Roychoudhury. (2026). “AutoCodeRover: Agentic Program Repair for SonarQube Issues.” *Industry Track, Symposium on the Foundations of Software Engineering (FSE)*.
- D. Wolff, M. Mirchev, A. Roychoudhury. (2026). “Large Language Models in Software Security Analysis.” *Communications of the ACM*, 69(6).
- Y. Zhang, J. Wang, D. Berzin, M. Mirchev, A. Roychoudhury. (2026). “Fixing Security Vulnerabilities with Agentic AI in OSS-Fuzz.” *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (ICSE)*.
- Y. Liu, Y. Song, M. Mirchev, A. Roychoudhury. (2025). “Computation Tree Logic Guided Program Repair.” *IEEE Transactions on Software Engineering*.
- R. Shariffdeen, B. Hassanshahi, M. Mirchev, A. El Husseini, et al. (2025). “Detecting Python Malware in the Software Supply Chain with Program Analysis.” *Proceedings of the 47th IEEE/ACM International Conference on Software Engineering (ICSE)*.
- M. Mirchev, A. Costea, A. K. Singh, A. Roychoudhury. (2024). “Assured Automatic Programming via Large Language Models.” *arXiv preprint*.
- R. Shariffdeen, Y. Noller, M. Mirchev, H. Ruan, G. Xiang, A. Costa, G. J. Duck, et al. (2024). “APR Competition 2024.” *Proceedings of the 5th ACM/IEEE International Workshop on Automated Program Repair*.
- R. Meng, M. Mirchev, M. Böhme, A. Roychoudhury. (2024). “Large Language Model guided Protocol Fuzzing.” *Network and Distributed System Security Symposium (NDSS)*.
- Z. Fan, X. Gao, M. Mirchev, A. Roychoudhury, S. H. Tan. (2023). “Automated Repair of Programs from Large Language Models.” *Proceedings of the 45th International Conference on Software Engineering (ICSE), Research Track*.
- R. Shariffdeen, M. Mirchev, Y. Noller, A. Roychoudhury. (2023). “Cerberus: A Program Repair Framework.” *Proceedings of the 45th International Conference on Software Engineering (ICSE), Tool Demonstrations Track*.
- R. Shariffdeen, M. Mirchev, A. Roychoudhury. (2023). “Program Repair Competition.” *2023 IEEE/ACM International Workshop on Automated Program Repair (APR)*.

Teaching

- Teaching Assistant, CS3234 Logic for Proofs and Programs – National University of Singapore, School of Computing (Semester 2, AY2025/2026)
- Teaching Assistant, CS3213 Foundations of Software Engineering – National University of Singapore, School of Computing (Semester 2, AY2022/2023)
- Teaching Assistant, Eindhoven University of Technology – 2ITB0 Provable Programming, 2IT80 Discrete Structures, 2IT60S Logic and Set Theory, 2IT90 Automata, Language Theory and Complexity

Service

- 2022: Student Volunteer, ESEC/FSE 2022
- Reviewer, IEEE Transactions on Software Engineering (TSE)
- Reviewer, Computers & Security (Elsevier)